

Smart Home Media Server

Ryan Arnzen

Project Summary

- Built a self-hosted home server on Ubuntu Server
- Centralized media streaming, cloud storage, and monitoring
- Used Docker to deploy and manage services
- Added Raspberry Pi integration for physical status indicators
- Designed the system to be expandable for automation and future smart-home features

Problem

Many families rely on multiple separate platforms for:

- media streaming
- file storage
- remote access

Problems

- repeat payments
- reduced privacy and control of data
- limited customization

Solution

My solution was to build a Docker-based smart home media platform with:

- Plex for local and remote media access

- Nextcloud for personal cloud storage
- Nginx Proxy Manager for reverse-proxy/domain access for both public-facing services
- Cloudflare Tunnel for domain-based external routing
- Prometheus + Grafana for monitoring
- node_exporter + cAdvisor for host and container metrics
- Raspberry Pi 4 for system status output

Architecture

Internet

- `cloud.arnzenserver.org` → Cloudflare Tunnel → Nginx Proxy Manager → Nextcloud
- `media.arnzenserver.org` → Cloudflare Tunnel → Nginx Proxy Manager → Plex

Ubuntu Server

- Plex
- Nextcloud
- Nginx Proxy Manager
- Cloudflared Tunnel
- Prometheus
- Grafana
- node_exporter
- cAdvisor

Raspberry Pi 4

- Python LED status script
- Physical server-health indicators

Technology Stack

- Ubuntu Server
- Docker / Docker Compose
- Plex

- Nextcloud
- Nginx Proxy Manager
- Prometheus
- Grafana
- node_exporter
- cAdvisor
- Raspberry Pi 4
- Python 3
- gpiozero

What Works

- Plex local access works
- Plex external access works through `media.arzenserver.org`
- Nextcloud access works through `cloud.arzenserver.org`
- Cloudflare Tunnel routing is working
- Nginx Proxy Manager routing is working
- Prometheus is collecting metrics
- Grafana dashboards are working
- Host and container monitoring are verified
- Raspberry Pi LED hardware wiring is working and ready for demo



Home

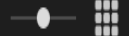
Home

Trending

Activity

Find Friends

My Profile



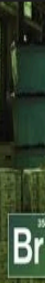
Continue Watching

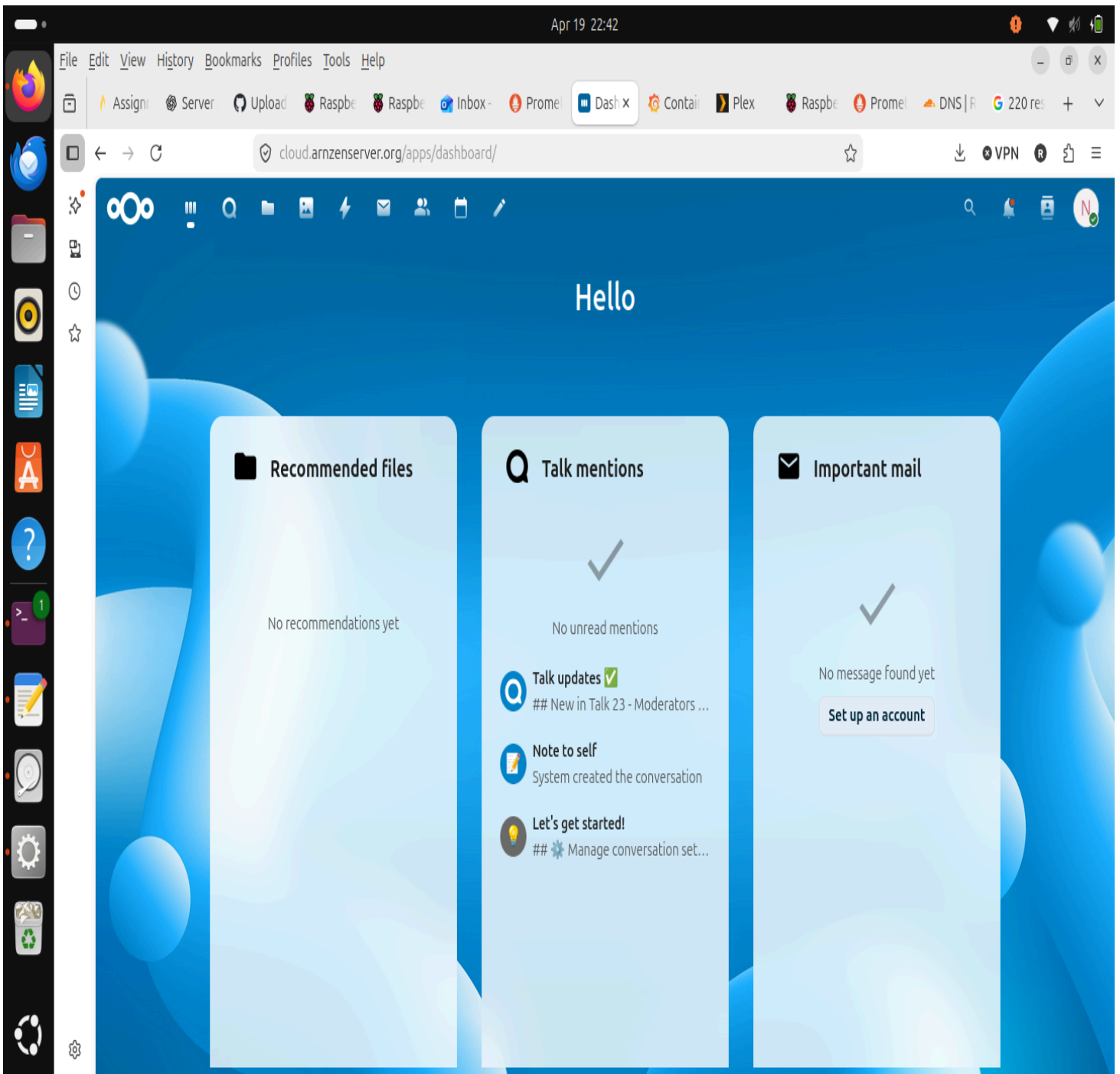


Mafia Wars

What's On Now

Live TV





Domain and Service Access

- Public domain routing is configured through Cloudflare
- External access is forwarded through Cloudflare Tunnel
- Nginx Proxy Manager routes traffic to internal services
- Plex and Nextcloud are both reachable through their own subdomains

←→↻

dash.cloudflare.com/478d47fb31a94018cf8dc1c1c3feff29/arnzenserver.org/dns/records

☆

↓VPNⓇ

📄

☰

🔥

Ryan.arnzen@outlook.co...

arnzenserver.org ⚙️ ☆ Free

🔮 Ask AI

🗣️ Support

👤

🔍 Quick search... Ctrl K

↩️ Back to Domains

📄 Overview

🕒 Recents >

🛡️ AI Crawl Control >

📄 Log Explorer >

🕒 Analytics & logs >

👤 DNS ▾

Records

Analytics

Settings

✉️ Email >

🔒 SSL/TLS >

🛡️ Security >

🔌 Access >

⚡ Speed >

📄 Caching >

📄

DNS management for
arnzenserver.org

Review, add, and edit DNS records. Changes propagate globally within minutes.

DNS Setup: Full ⓘ [Import and Export](#) ▾ ⚙️ [Dashboard Display Settings](#)

Search DNS Records

🔍 Add filter

🔍

🔍 Search

➕ Add record

☐	Type ⓘ	▲	Name ⓘ	Content ⓘ	Proxy status ⓘ	TTL ⓘ	Actions
☐	Tunnel		cloud	homelab	🔥 Proxied	Auto	Edit ▶
☐	Tunnel		media	homelab	🔥 Proxied	Auto	Edit ▶

Cloudflare Nameservers

Every DNS zone on Cloudflare is assigned a set of Cloudflare-branded nameservers.

Type	Value
NS	giancarlo.ns.cloudflare.com
NS	meiling.ns.cloudflare.com

Monitoring

Monitoring services

- Prometheus for metrics collection
- Grafana for dashboards
- node_exporter for host metrics
- cAdvisor for container metrics

What is monitored

- CPU usage

- RAM usage
- disk usage
- uptime
- container memory
- container CPU
- container count

Prometheus Verification

- Prometheus targets are up
- cAdvisor is being scraped
- node_exporter is being scraped
- Prometheus is scraping its own metrics
- Monitoring pipeline is functioning correctly

← → ↻ Not Secure http://192.168.1.11:9090/targets ☆

⬇️ VPN ⓘ 📄 ☰

Prometheus 🔍 Query 🔔 Alerts 🏠 Status > Target health ▾ 🌙 ⚙️ 📖

Select scrape pool ⌵ 🏠 Filter by target health ⌵ 🔍 Filter by endpoint or labels 📄

cadvisor 1 / 1 up 🟢 ^			
Endpoint	Labels	Last scrape	State
http://cadvisor:8080/metrics	instances="cadvisor:8080" job="cadvisor" ▾	🕒 15.572s ago 🕒 178ms	UP

node_exporter 1 / 1 up 🟢 ^			
Endpoint	Labels	Last scrape	State
http://node-exporter:9100/metrics	instances="node-exporter:9100" job="node_exporter" ▾	🕒 27.355s ago 🕒 225ms	UP

prometheus 1 / 1 up 🟢 ^			
Endpoint	Labels	Last scrape	State
http://prometheus:9090/metrics	instances="prometheus:9090" job="prometheus" ▾	🕒 9.728s ago 🕒 10ms	UP

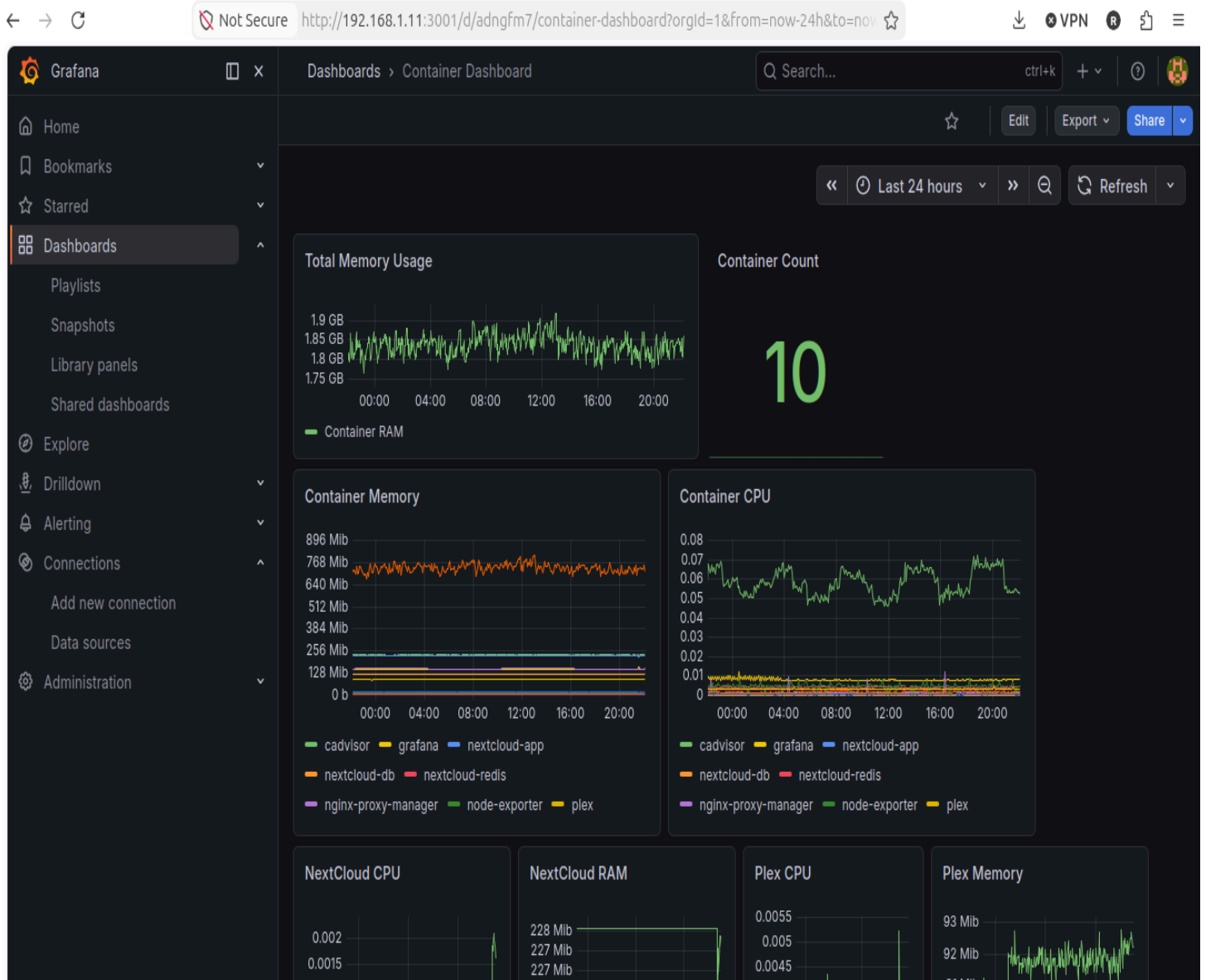
Grafana Results

Node Exporter dashboard shows

- disk usage
- root filesystem usage
- uptime
- total root filesystem size

Container dashboard shows

- total memory usage
- container count
- per-container CPU usage
- per-container memory usage
- Plex CPU and memory
- Nextcloud CPU and memory



Raspberry Pi

Current design

- Pi polls health and monitoring data
- Python script controls LEDs
- LEDs represent server conditions

LED mapping

- Green = server reachable
- Blue = Plex service healthy

- Yellow = resource warning
- Red = critical issue / disk or service problem

LED Hardware Setup

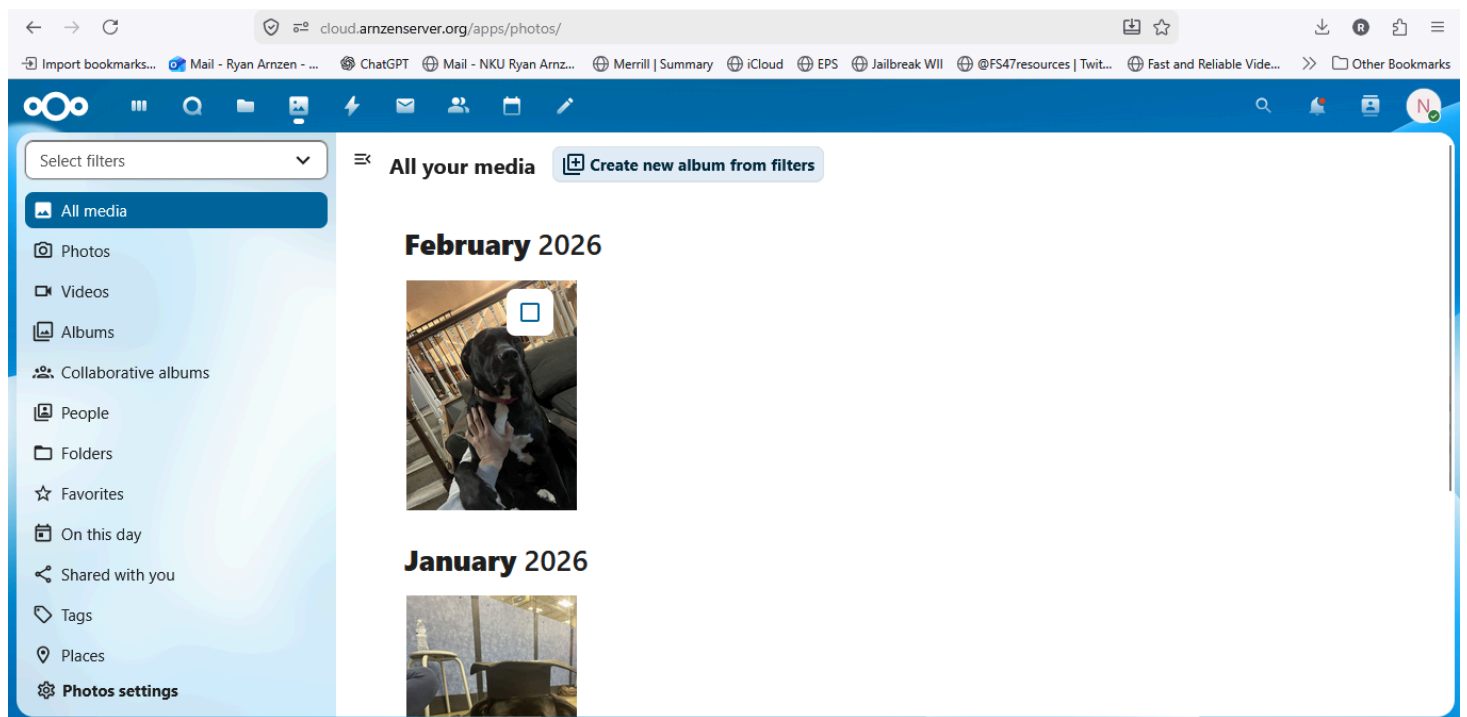
Wired outputs

- GPIO17 → Green
- GPIO27 → Blue
- GPIO19 → Yellow
- GPIO26 → Red

Circuit pattern

- GPIO → resistor → LED anode
- LED cathode → GND

Demo Video



Learning With AI

AI was used to:

- assist research
- help troubleshooting
- help document and track changes
- help create Python code

Workflow

Ask AI for guidance → ask questions → implement manually → test the system → document the verified result

AI mainly helped with:

- Docker/Linux troubleshooting
- reverse proxy design
- monitoring setup
- Raspberry Pi coding

Challenges and Fixes

Challenges

- getting remote access correct
- understanding reverse proxy vs direct service access
- debugging Grafana “No data” panels
- deciding between LCD display and LEDs
- Raspberry Pi reflash / setup issues

Fixes

- validated Plex remote access directly
- kept Nextcloud on reverse proxy path
- confirmed Prometheus targets and Grafana datasource
- switched to simpler LED hardware strategy
- rebuilt the Pi from a clean flash

Future Improvements

- extend Raspberry Pi from LEDs to LCD/OLED display
- add alerts and notifications
- expand monitoring into predictive maintenance
- potentially connect with the separate ML monitoring project later

Questions?